

ASYMPTOTIC DISTRIBUTIONS OF CERTAIN TEST CRITERIA OF NORMALITY

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1. Introduction

Let $X_1, X_2, \dots, X_n$ be independent and identically distributed random variables having the common distribution function F . Let HS denote the hypothesis that $F = \Phi$, the standard normal distribution function and let HC denote the hypothesis that F is some normal distribution, i.e. $F(x) = \Phi((x-\mu)/\sigma)$ for some unspecified μ and σ . Further, let $X_{1n}, \dots, X_{nn}$ denote the order statistics corresponding to $X_1, \dots, X_n$ and put $H = \Phi^{-1}$ and $\phi = \Phi'$.

Under HS the expectation of X_{in} is approximately $H(i/n+1) \equiv H_{in}$; hence an intuitively reasonable statistic for testing HS is

$$(1) \quad L_n^0 = \sum (X_{in} - H_{in})^2.$$

Let $\bar{X}_n = n^{-1} \sum X_i$, $s_n^2 = n^{-1} \sum (X_i - \bar{X}_n)^2$; standardizing X_{in} by subtracting $\bar{X}_n$ and dividing by s_n , a statistic for testing HC , related to (1), is

$$(2) \quad L_n = \sum [(X_{in} - \bar{X}_n)/s_n - H_{in}]^2.$$

This statistic can also be justified intuitively thus (see also van der Watt (1969)): Under HC $EX_{in} \approx \mu + \sigma H_{in}$ for n large; so the regression line of X_{in} on H_{in} should be approximately linear under HC ; the fraction of "variation explained" is then r_n^2 with

$$(3) \quad r_n = n^{-1} \sum X_{in} H_{in} / s_n t_n$$

and $t_n^2 = n^{-1} \sum H_{in}^2$, and r_n^2 should be large; as $\sigma > 0$, r_n should also be positive under HC ; this suggests rejecting HC if $r_n < 1 - c$ for suitable c . Squaring the summands in (2) and simplifying lead to

$$(4) \quad L_n = 2nt_n(1 - r_n) + n(1 - t_n)^2$$

so that rejecting for small r_n is equivalent to rejecting for large L_n .

The statistic W' of Shapiro and Francia (1972) is closely related to r_n^2 ; W' differs from r_n^2 essentially just in having H_{in} replaced by EX_{in} under HS so that r_n^2 could be thought of as a large sample approximation to W' .

The object of this paper is to obtain the asymptotic distributions of L_n^0 and L_n under the relevant hypotheses. In section 2 some preliminary results are given; sections 3 and 4 contain characterizations of the asymptotic distributions of L_n^0 and L_n respectively; from these the characteristic functions can be calculated and section 5 shows how these can be inverted; these distributions together with their percentiles are then tabled.

2. Preliminaries

Symbols R_n , R_{1n} , etc. will be used to indicate terms tending to zero in a suitable sense as $n \rightarrow \infty$; unless otherwise indicated all limits will be with respect to $n \rightarrow \infty$; $A_1, A_2, \dots$, will denote finite fixed constants and for $c(x)$ a function on $(0,1)$ we abbreviate $c(i/n+1)$ by c_{in} ; similarly $c_{ijn} \equiv c(i/n+1, j/n+1)$ for $c(x, y)$ a function on the unit square. The notation introduced in the previous paragraph will be used below without further clarification. Finally $[z]$ denotes the largest integer less than or equal to z .

Lemma 1. If $b_n = H(b/n)$ for some fixed $b > 0$, then $|b_n| \phi(b_n) = O(n^{-1} \log n)$ and $b_n^2 = O(\log n)$.

Proof: From the well known asymptotic relation $\Phi(z) \sim \phi(z)/|z|$ for $z \rightarrow -\infty$ we get $n = b/\Phi(b_n) \sim b/|b_n| \phi(b_n)$ and $\log n \sim b_n^2/2$; hence $n|b_n| \phi(b_n) \log n \sim 2b$.

Lemma 2. $t_n^2 = 1 + O(n^{-1} \log n)$.

Proof: We have

$$\begin{aligned} t_n^2 - 1 &= n^{-1} \sum H_{in}^2 - \int_0^1 H(x)^2 dx \\ &= R_{1n} + R_{2n} + R_{3n} \end{aligned}$$

where

$$\begin{aligned} R_{1n} &= n^{-1} \sum_{i \leq [n/2]} H_{in}^2 - \int_0^{[n/2]/n} H(x)^2 dx \\ R_{2n} &= n^{-1} \sum_{i \geq [n/2]+1} H_{in}^2 - \int_{([n/2]+1)/n}^1 H(x)^2 dx \\ R_{3n} &= - \int_{[n/2]/n}^{([n/2]+1)/n} H(x)^2 dx. \end{aligned}$$

Clearly $R_{3n} = o(n^{-1})$. Also using monotonicity of $H(x)^2$ for $0 < x < 1/2$, we get

$$|R_{1n}| \leq \sum_{i \leq [n/2]} \int_{(i-1)/n}^{i/n} |H_{in}^2 - H(x)^2| dx \leq \sum_{2 \leq i \leq [n/2]} \int_{(i-1)/n}^{i/n} [H(i-1/n)^2 - H(i/n)^2] dx \\ + n^{-1} H_{1n}^2 + \int_0^{1/n} H(x)^2 dx \leq n^{-1} H(1/n)^2 - n^{-1} H([n/2]/n)^2 + n^{-1} H_{1n}^2 + \int_{-\infty}^{H(1/n)} x^2 \phi(x) dx$$

The second term on the right is $o(n^{-1})$; the first and third terms are both $O(n^{-1} \log n)$ by lemma 1. The last term equals $n^{-1} - H(1/n)\Phi(H(1/n))$ by partial integration and this is also $O(n^{-1} \log n)$ by lemma 1. Hence $R_{1n} = O(n^{-1} \log n)$. R_{2n} can be handled similarly.

Lemma 3. Let $U_{1n}, \dots, U_{nn}$ denote the order statistics of an independent sample from the uniform distribution over $(0, 1)$; with $V_{in} = U_{in} - i/n + 1$ we then have

$$EV_{in}^2 = v_{in}/n + 2$$

and

$$EV_{in}^4 < 27 v_{in}^2/n^2 \text{ for } n \geq 5$$

where

$$v(x) = x(1-x).$$

Proof: It is well known that U_{in} has a Beta distribution with parameters i and $n-i+1$ and the results follow from straightforward calculations.

Lemma 4. Let U_{in} be as above and let U_{in}^* be between U_{in} and $i/n + 1$, $i = 1, \dots, n$. Define

$$R_n(\alpha, \beta, \gamma, \tau) = \sum_{i=1}^n V_{in}^{\alpha} H_{in}^{\beta} H_{in}^{\gamma} H_{in}^{\tau} / (U_{in}^*)^{\tau}$$

with

$$0 < \alpha \leq 4, \beta \geq 0, \gamma \geq 0 \text{ and } \tau \geq 0.$$

Let $\Delta = \alpha/2 - \gamma - 2\tau$. Then $R_n(\alpha, \beta, \gamma, \tau)$ is

$$(a1) \quad o_p(n^{1-\alpha/2} \log n) \text{ if } \beta = \gamma + \tau, \Delta > -1$$

$$(a2) \quad o_p(n^{1-\alpha/2} (\log n)^2) \text{ if } \beta = \gamma + \tau, \Delta = -1$$

$$(a3) \quad o_p(n^{-\Delta-\alpha/2} \log n) \text{ if } \beta = \gamma + \tau, \Delta < -1$$

- (b1) $O_p(n^{1-\alpha/2} \log n)$ if $\beta < \gamma + \tau$, $\Delta > -1$
 (b2) $O_p(n^{1-\alpha/2} \log n)$ if $\beta = \gamma + \tau$, $\Delta = -1$
 (b3) $O_p(n^{-\Delta-\alpha/2})$ if $\beta < \gamma + \tau$, $\Delta < -1$.

Proof. Let $\epsilon > 0$; we apply proposition 4 of Chernoff, Gastwirth and Johns (1967); in their notation, define

$$(5) \quad \mu_{in} \equiv \mu_{in}(\epsilon) = \sup_{u_{in}(\epsilon) \leq u \leq u_{in}(\epsilon)} |H''(u)|.$$

Then, with probability at least $1 - \epsilon$, we have

$$(6) \quad |R_n(\alpha, \beta, \gamma, \tau)| \leq \Sigma |V_{in}|^\alpha |H_{in}|^\beta |H'_{in}|^\gamma \mu_{in}^\tau.$$

Now

$$E|V_{in}|^\alpha \leq (E V_{in}^4)^{\alpha/4} \leq A_1 n^{-\alpha/2} \nu_{in}^{\alpha/2}$$

by lemma 3. The expected value of the righthand side of (6) is therefore no more than

$$(7) \quad A_1 n^{-\alpha/2} \Sigma \nu_{in}^{\alpha/2} |H_{in}|^\beta |H'_{in}|^\gamma \mu_{in}^\tau.$$

Choose and fix δ with $0 < \delta < 1/2$. Split (7) into three parts, viz. $A_1 R_{1n}(\delta)$ in which summation extends over the indices 1 to $[n\delta]$, $A_1 R_{2n}(\delta)$ in which summation is over $[n\delta] + 1$ to $[n(1-\delta)] - 1$ and $A_1 R_{3n}(\delta)$ being the remainder.

then
$$\text{If } A_2(\delta) = \sup_{[n\delta]/n+1 \leq u \leq [n(1-\delta)]/n+1} |H''(u)|,$$

$$R_{2n}(\delta) \leq A_2(\delta)^\tau n^{1-\alpha/2} \Sigma \nu_{in}^{\alpha/2} |H_{in}|^\beta |H'_{in}|^\gamma$$

and since the $\nu(x)^{\alpha/2} |H(x)|^\beta |H'(x)|^\gamma$ is bounded and continuous over $[\delta, 1-\delta]$

$$n^{-1} \Sigma \nu_{in}^{\alpha/2} |H_{in}|^\beta |H'_{in}|^\gamma \rightarrow \int_0^{1-\delta} \nu(x)^{\alpha/2} |H(x)|^\beta |H'(x)|^\gamma dx < \infty$$

Hence

$$(8) \quad R_{2n}(\delta)/n^{1-\alpha/2} \log n \rightarrow 0.$$

Regarding $R_{1n}(\delta)$ first note that $u_{in}(\epsilon) < 1/2$ for $i \leq [n\delta]$ and n large enough; then $\mu_{in}(\epsilon) = |H''(u_{in}(\epsilon))|$ as $|H''(x)|$ is decreasing over $(0, 1/2)$. Abbreviate $u_{in}(\epsilon)$ by u_{in} and $u_{in}(\epsilon)$ by

u_{in} ; since $H''(x) = H'(x)^2 H(x)$, the i -th term of $R_{1n}(\delta)$ is no more than

$$(9) \quad n^{-\alpha/2} (i/n+1)^{\alpha/2} |H_{in}|^{\beta} |H'_{in}|^{\gamma} |H(u_{in})|^{\tau} |H'(u_{in})|^{2\tau} \\ \leq n^{-\alpha/2} [(i/n+1) |H'_{in}|^{\gamma} |u_{in} H'(u_{in})|^{2\tau} |H_{in}|^{\beta} |H(u_{in})|^{\tau} (i/n+1)^{\alpha/2 - \gamma} u_{in}^{-2\tau}].$$

From the inequality $\Phi(z) \leq \phi(z)/|z|$ for $z < 0$ follows $x H'(x) \leq |H(x)|^{-1}$ for $0 < x < 1/2$; hence (9) is bounded by

$$(10) \quad n^{-\alpha/2} |H_{in}|^{\beta - \gamma} |H(u_{in})|^{-\tau} (i/n+1)^{\alpha/2 - \gamma - 2\tau} (u_{in}^{in}/u_{in})^{2\tau}$$

noting that $i/n+1 \leq u_{in}^{in}$. From this and $u_{in}^{in}/u_{in} \leq K_1(\epsilon)$ follows that (10) is bounded by

$$(11) \quad K_1(\epsilon)^{2\tau} n^{-\alpha/2} |H_{in}|^{\beta - \gamma - \tau} (i/n+1)^{\alpha/2 - \gamma - 2\tau}.$$

In case (a1) $\beta = \gamma + \tau$ so $R_{1n}(\delta)$ is bounded by

$$(12) \quad K_1(\epsilon)^{2\tau} n^{-\alpha/2} (n+1)^{-\alpha/2 + \gamma + 2\tau} \sum_{i \leq [n\delta]} i^{\alpha/2 - \gamma - 2\tau} \\ \sim A_3 K_1(\epsilon)^{2\tau} n^{-\alpha + \gamma + 2\tau} (n\delta)^{\alpha/2 - \gamma - 2\tau + 1} \\ = A_3 K_1(\epsilon)^{2\tau} \delta^{\alpha/2 - \gamma - 2\tau + 1} n^{1 - \alpha/2}.$$

Hence $R_{1n}(\delta) = o(n^{1 - \alpha/2} \log n)$; a similar result holds for $R_{3n}(\delta)$ and this together with (8) shows that (7) is $o(n^{1 - \alpha/2} \log n)$ and from (6) (a1) follows.

In case (a2) (12) must be replaced by

$$K_1(\epsilon)^{2\tau} n^{-\alpha/2} (n+1)^{-\alpha/2 + \gamma + 2\tau} \sum_{i \leq [n\delta]} i^{-1} \sim A_4 K_1(\epsilon)^{2\tau} n^{1 - \alpha/2} \log n$$

so that $R_{1n}(\delta) = o(n^{1 - \alpha/2} (\log n)^2)$ and again a similar result holds for $R_{3n}(\delta)$; (a2) then follows.

In case (a3) (12) must be replaced by

$$K_1(\epsilon)^{2\tau} n^{-\alpha/2} (n+1)^{-\alpha/2 + \gamma + 2\tau} \sum_{i \leq [n\delta]} i^{\alpha/2 - \gamma - 2\tau} \sim A_5 K_1(\epsilon)^{2\tau} n^{-\alpha + \gamma + 2\tau}$$

Hence $R_{1n}(\delta) = o(n^{-\alpha + \gamma + 2\tau} \log n)$ and since $1 - \alpha/2 < -\alpha + \gamma + 2\tau$ from (8) follows that also $R_{2n}(\delta) = o(n^{-\alpha + \gamma + 2\tau} \log n)$. Again a similar result holds for $R_{3n}(\delta)$ and it follows that (a3) is true. Case (b1) is treated exactly like (a1).

For the case (b2) $|H_{in}|^{\beta-\gamma-\tau} \leq |H(\delta)|^{\beta-\gamma-\tau}$ for $i \leq [n\delta]$; hence from (11) $R_{1n}(\delta)$ is bounded by

$$K_1(\epsilon)^{2\tau} |H(\delta)|^{\beta-\gamma-\tau} n^{-\alpha/2} (n+1)^{-\alpha/2+\gamma+2\tau} \sum_{i \leq [n\delta]} i^{-1} \\ \sim A_6 K_1(\epsilon)^{2\tau} |H(\delta)|^{\beta-\gamma-\tau} n^{1-\alpha/2} \log n.$$

So $\lim_{\delta \rightarrow 0} \limsup_{n \rightarrow \infty} R_{1n}(\delta)/n^{1-\alpha/2} \log n = 0$. From (8) we get the same result with $R_{1n}(\delta)$ replaced by $R_{2n}(\delta)$ and a similar result holds for $R_{3n}(\delta)$ by symmetry. Hence (7) tends to zero as $n \rightarrow \infty$ and from (6) (b2) follows. (b3) is obtained in a similar manner.

3. The Asymptotic Distribution of L_n^0

Define

$$(13) \quad a_n^0 = (n+1)^{-1} \sum_{k=1}^n h_{kn} \nu_{kn}$$

with $h(x)$ given in (15) below.

Theorem 1. Under HS we have

$$D(L_n^0 - a_n^0) \rightarrow D(\sum_1^\infty (Y_m^2 - 1)/m)$$

where $Y_1, Y_2, \dots$ are independent and identically distributed $N(0,1)$ random variables.

Proof. Let $U_{in} = \Phi(X_{in})$; then $U_{1n}, \dots, U_{nn}$ are distributed as the order statistics of an independent sample of size n from the uniform distribution over $(0,1)$; then $X_{in} = H(U_{in})$ and by Taylor's expansion,

$$(14) \quad X_{in} = H_{in} + V_{in} H'_{in} + \frac{1}{2} V_{in}^2 H''(U_{in}^*)$$

with U_{in}^* between U_{in} and $i/n+1$. Put

$$(15) \quad h(x) = H'(x)^2.$$

Substituting into L_n^0 , we get

$$(16) \quad L_n^0 = Q_n^h + R_{4n} + R_{5n}$$

where

$$(17) \quad Q_n^h = \sum V_{in}^2 h_{in}$$

and

$$(18) \quad R_{4n} = \Sigma V_{in}^3 H'_{in} H''(U_{in}^*)$$

$$(19) \quad R_{5n} = \frac{1}{4} \Sigma V_{in}^4 H''(U_{in}^*)^2.$$

In the notation of lemma 4 we have $R_{4n} = R_n(3,0,1,1)$ which corresponds to case (b3); hence $R_{4n} = o_p(1)$. Also $R_{5n} = R_n(4,0,0,2)$ which again falls under case (b3); hence $R_{5n} = o_p(1)$.

De Wet (1972), (see also De Wet and Venter (1972)) proved that

$$D(Q_n^h - a_n^0) \rightarrow D(\Sigma_1^\infty (Y_m^2 - 1)/m).$$

From (16) and this fact the theorem follows.

4. The Asymptotic Distribution of L_n

Define

$$(20) \quad a_n = a_n^0 - a'_n - a''_n$$

with a_n^0 as given by (13) and

$$(21) \quad a'_n = n^{-1}(n+1)^{-1} \sum_{i,j=1}^n H'_{in} H'_{jn} T_{ijn}$$

$$(22) \quad a''_n = n^{-1}(n+1)^{-1} \sum_{i,j=1}^n H'_{in} H_{in} H'_{jn} H_{jn} T_{ijn}$$

where

$$(23) \quad T(x,y) = \begin{cases} x(1-y) & \text{for } x \leq y \\ y(1-x) & \text{for } x > y \end{cases}$$

Theorem 2. Under HC we have

$$(24) \quad D(2n(1-r_n) - a_n) \rightarrow D(\Sigma_{m=3}^\infty (Y_m^2 - 1)/m)$$

$$(25) \quad D(L_n - a_n) \rightarrow D(\Sigma_{m=3}^\infty (Y_m^2 - 1)/m).$$

Proof. Since the distribution of L_n is invariant under scale and translation changes it involves no loss of generality to assume $\mu=0$ and $\sigma=1$.

From (14) and (3) we get

$$(26) \quad r_n = (t_n^2 + w_n + v_n)/s_n t_n$$

where

$$(27) \quad w_n = n^{-1} \sum V_{in} H_{in} H'_{in}$$

$$(28) \quad v_n = n^{-1} \sum e_{in} H_{in}$$

with

$$(29) \quad e_{in} = \frac{1}{2} V_{in}^2 H''(U_{in}^*).$$

Also from (14),

$$(30) \quad \bar{X}_n = y_n + e_n$$

with

$$(31) \quad y_n = n^{-1} \sum V_{in} H'_{in}, \quad e_n = n^{-1} \sum e_{in}.$$

Further

$$(32) \quad s_n^2 = n^{-1} \sum X_{in}^2 - \bar{X}_n^2 \\ = t_n^2 + 2w_n + 2v_n + n^{-1} Q_n^h - y_n^2 + R_{6n}$$

where

$$(33) \quad R_{6n} = n^{-1} \sum e_{in}^2 + 2n^{-1} \sum e_{in} V_{in} H'_{in} - e_n^2 - 2e_n y_n.$$

From (26) and (32) we get

$$(34) \quad s_n^2 t_n^2 (1 - r_n^2) = t_n^2 (n^{-1} Q_n^h - y_n^2 - w_n^2) + R_{7n}$$

where

$$(35) \quad R_{7n} = R_{6n} t_n^2 + w_n^2 (t_n^2 - 1) - v_n^2 - 2v_n w_n.$$

We now show that

$$(36) \quad n R_{7n} = o_p(1).$$

The first term of nR_{6n} is $R_n(4,0,0,2) = o_p(1)$ by lemma 4; the second term of nR_{6n} is $R_n(3,0,1,1) = o_p(1)$. Also $ne_n = R_n(2,0,0,1) = o_p(\log n)$ so that $ne_n^2 = o_p(n^{-1}(\log n)^2) = o_p(1)$. Similarly the remainder of the terms can be handled by lemmas 2 and 4.

We now use the fact that

$$(37) \quad V_{kn} = S_{n+1}^{-1} \sum_{i=1}^{n+1} \psi_{ikn} Z_i$$

where $Z_1 + 1, Z_2 + 1, \dots$ are independent exponentially distributed with expectation 1 and

$$(38) \quad \psi(x, y) = \begin{cases} 1 - y & \text{for } x \leq y \\ -y & \text{for } x > y \end{cases}$$

and

$$(39) \quad S_{n+1} = \sum_{i=1}^{n+1} (Z_i + 1)$$

(see De Wet (1972) and De Wet and Venter (1972)). Substituting into y_n and changing the order of summation, we get

$$(40) \quad ny_n^2 = (n+1/S_{n+1})^2 T'_{n+1}$$

with

$$(41) \quad T'_{n+1} = (n+1)^{-1} \sum_{ij} c'_{ijn+1} Z_i Z_j$$

$$(42) \quad c'_{ijn+1} = n^{-1}(n+1)^{-1} \sum_{k,m=1}^n H'_{kn} H'_{mn} \psi_{ikn} \psi_{jmn}.$$

Similarly

$$(43) \quad nw_n^2 = (n+1/S_{n+1})^2 T''_n$$

with

$$(44) \quad T''_{n+1} = (n+1)^{-1} \sum_{i,j=1}^{n+1} c''_{ijn+1} Z_i Z_j$$

$$(45) \quad c''_{ijn+1} = n^{-1}(n+1)^{-1} \sum_{k,m=1}^n H_{kn} H'_{kn} H_{mn} H'_{mn} \psi_{ikn} \psi_{jmn}.$$

De Wet and Venter (1972) proved that

$$(46) \quad Q_n^h = (n+1/S_{n+1})^2 T_{n+1}^o$$

with

$$(47) \quad T_{n+1}^0 = (n+1)^{-1} \sum_{i,j=1}^{n+1} c_{ijn+1}^0 Z_i Z_j$$

$$(48) \quad c_{ijn+1}^0 = (n+1)^{-1} \sum_{k=1}^n h_{kn} \psi_{ikn} \psi_{jkn}.$$

Hence

$$(49) \quad Q_n^h - n y_n^2 - n w_n^2 = (n+1/S_{n+1})^2 T_{n+1}$$

with

$$(50) \quad T_{n+1} = (n+1)^{-1} \sum_{i,j=1}^{n+1} c_{ijn+1} Z_i Z_j$$

$$(51) \quad c_{ijn} = c_{ijn}^0 - c'_{ijn} - c''_{ijn}.$$

In order to get the asymptotic distribution of T_n we apply theorem 1 of De Wet and Venter (1972); we proceed as in their paragraph 3 and set

$$(52) \quad c(x,y) = c^0(x,y) - c'(x,y) - c''(x,y)$$

with

$$(53) \quad c^0(x,y) = \int_0^1 h(z) \psi(x,z) \psi(y,z) dz$$

$$(54) \quad c'(x,y) = \int_0^1 \int_0^1 H'(u) H'(v) \psi(x,u) \psi(y,v) du dv \\ = g_1(x) g_1(y)$$

$$(55) \quad c''(x,y) = \int_0^1 \int_0^1 H(u) H'(u) \psi(x,u) H(v) H'(v) \psi(y,v) du dv \\ = \frac{1}{2} g_2(x) g_2(y)$$

where

$$(56) \quad g_m(x) = H_m(H(x)), \quad m = 1, 2, \dots$$

with H_m the m -th normalized Hermitian polynomial. With $\gamma_m = 1/m$ it follows from De Wet and Venter (1972) that $\{g_1, g_2, \dots\}$ and $\{1/2, \dots\}$ are the eigenfunctions and values of $c^0(x,y)$ and consequently from (52) and (54), (55) that $\{g_3, g_4, \dots\}$ and $\{1/3, 1/4, \dots\}$ are

the eigenfunctions and values of c . This means that we choose

$$(57) \quad \begin{aligned} b_{imn} &= g_m(i/n+1) \\ \gamma_m &= m^{-1} \end{aligned} \quad m = 3, 4, \dots$$

and verify the conditions of theorem 1 of De Wet and Venter (1972). Details of this can be found in De Wet (1972). Thus

$$(58) \quad D(T_{n+1} - ET_{n+1}) \rightarrow D(\Sigma_m^\infty = 3(Y_m^2 - 1)/m).$$

But $ET_{n+1} = a_n$ as can be verified from (20), (21) and (22) together with (41)–(45). From (21) and (22)

$$a'_n \rightarrow \int_0^1 \int_0^1 H'(x)H'(h)T(x,y)dx dy = 1$$

$$a''_n \rightarrow \int_0^1 \int_0^1 H(x)H'(x)H'(y)H(y)T(x,y)dx dy = 1/2.$$

It was shown in (3) that $a_n^0 = o(\log n)$; hence also $a_n = o(\log n)$. From (49)

$$(59) \quad Q_n^h - ny_n^2 - nw_n^2 - a_n = (n+1/S_{n+1})^2(T_{n+1} - a_n) + o_p(1)$$

using the fact that $n+1/S_{n+1} = 1 + o_p(n^{-1/2})$ by the central limit theorem. By (34) and (59)

$$(60) \quad s_n^2 t_n^2 [n(1-r_n^2) - a_n] = t_n^2 (n+1/S_{n+1})(T_{n+1} - a_n) + o_p(1).$$

Since $s_n^2 t_n^2 \rightarrow 1$, $n+1/S_{n+1} \rightarrow 1$ in probability it follows that $n(1-r_n^2) - a_n$ has the same asymptotic distribution as $T_{n+1} - ET_{n+1}$ given in (58).

It follows easily from (26) that $r_n \rightarrow 1$ in probability; hence

$$2n(1-r_n) - a_n = n(1-r_n^2) - a_n + n(1-r_n)^2.$$

Now $n^{1/2}(1-r_n^2) = o_p(1)$, i.e.

$$n^{1/2}(1-r_n)(1+r_n) = o_p(1) \text{ or } n^{1/2}(1-r_n) = o_p(1);$$

i.e. $n(1-r_n)^2 = o_p(1)$; hence $2n(1-r_n) - a_n$ also has the same asymptotic distribution so that (24) follows. (25) follows from (4).

Remark. Comparing the asymptotic distributions of L_n^0 and L_n we note the absence of the term $Y_1^2 + \frac{1}{2}Y_2^2$ in the latter; this phenomenon is similar to that of a loss of two degrees of freedom associated with a χ^2 -test for HC as compared with a χ^2 -test for HS.

5. Inversion of Characteristic Functions

Let f_o and f denote the characteristic functions of the asymptotic distributions of $L_n^O - a_n^O$ and $L_n - a_n$ respectively and let Ψ_O and Ψ be the corresponding distribution functions.

It follows that

$$(61) \quad f_o(t) = \prod_{k=1}^{\infty} e^{-it/k(1-2it/k)^{-1/2}}$$

and

$$(62) \quad f(t) = \prod_{k=3}^{\infty} e^{-it/k(1-2it/k)^{-1/2}}.$$

To invert these characteristic functions we need them in a different form.

Put: $f_o(t) = r_o(t)e^{i\theta_o(t)}$; then

$$(63) \quad r_o(t) = (\sinh 2\pi t/2\pi t)^{-1/4} \\ = (e^{2\pi t} - e^{-2\pi t}/4\pi t)^{-1/4}$$

and:

$$(64) \quad \theta_o(t) = \frac{1}{2} \sum_{k=1}^{\infty} (\tan^{-1} 2t/k - 2t/k).$$

Also, if $f(t) = r(t)e^{i\theta(t)}$, then:

$$(65) \quad r(t) = r_o(t)(1+4t^2)^{+1/4}(1+t^2)^{+1/4}$$

and

$$(66) \quad \theta(t) = \theta_o(t) - 1/2(\tan^{-1} 2t + \tan^{-1} t) + 3t/2.$$

From (63) and (65) it is obvious that $\int |f_o(t)|dt$, $\int |f(t)|dt < \infty$. Thus, from the inversion formula (Feller (1966), p. 483) it follows that:

$$(67) \quad \Psi_o(y) - \Psi_o(o) = 1/2\pi \int_{-\infty}^{\infty} (f_o(t)/t)(1 - e^{-ity})dt$$

and a similar expression for $\Psi(y) - \Psi(0)$ can be written down.

Substituting for $f_o(t)$ into (67) and taking real parts, yield:

$$(68) \quad \Psi_o(u) - \Psi_o(0) = 1/\pi \int_0^{\infty} (r_o(t)/t)(\sin \theta_o(t) + \sin(ty - \theta_o(t)))dt.$$

Similarly:

$$(69) \quad \Psi(y) - \Psi(0) = 1/\pi \int_0^{\infty} (r(t)/t)(\sin \theta(t) + \sin(ty - \theta(t)))dt.$$

The integrations in (68) and (69) can be carried out numerically. Taking y very large in (68) we can obtain $1 - \Psi_0(0)$ and thus $\Psi_0(y)$ for all y . Similarly for $\Psi(y)$. These two distribution functions are tabulated in tables 1 and 2 below.

Let $y_0(\alpha)$ and $y(\alpha)$ be such that $\Psi_0(y_0) = \Psi(y) = 1 - \alpha$. Table 3 then gives $y_0(\alpha)$ and $y(\alpha)$ for certain choices of α .

Table 4 gives the constants a_n^0 and a_n for certain choices of n .

From table 4 it follows that $a_n \approx a_n^0 - 3/2$. This also follows from the proof of theorem 2 where we have shown that $a_n' \rightarrow 1$ and $a_n'' \rightarrow 1/2$ whilst $a_n = a_n^0 - a_n' - a_n''$.

It follows from the proof of Theorem 2 that $n(1 - r_n^2) - a_n$ has the asymptotic distribution Ψ ; consequently approximate percentage points for r_n^2 can also be obtained from Table 3. For example, with $n = 100$ the 5% point for r_n^2 is $1 - (a_{100} + y(0,05))/100 = 1 - (1,15 + 1,63)/100 = 0,972$; the empirical 5% point for $n = 99$ for the W' test of Shapiro and Francia (1972) is 0,976 which compares well with the asymptotic value for r_n^2 .

TABLE 1

The distribution function $\Psi_0(y)$

y	$\Psi_0(y)$	y	$\Psi_0(y)$	y	$\Psi_0(y)$	y	$\Psi_0(y)$
-3,5	0,000	-0,8	0,381	1,5	0,836	3,8	0,960
-3,0	0,002	-0,7	0,410	1,6	0,846	3,9	0,963
-2,9	0,003	-0,6	0,438	1,8	0,856	4,0	0,965
-2,8	0,005	-0,5	0,466	1,8	0,864	4,5	0,974
-2,7	0,008	-0,4	0,494	1,9	0,873	5,0	0,981
-2,6	0,012	-0,3	0,520	2,0	0,880	5,5	0,986
-2,5	0,018	-0,2	0,546	2,1	0,887	6,0	0,989
-2,4	0,025	-0,1	0,570	2,2	0,894	6,5	0,992
-2,3	0,034	0,0	0,594	2,3	0,901	7,0	0,994
-2,2	0,045	0,1	0,617	2,4	0,907	7,5	0,996
-2,1	0,058	0,2	0,638	2,5	0,912	8,0	0,997
-2,0	0,073	0,3	0,659	2,6	0,917	8,5	0,998
-1,9	0,091	0,4	0,679	2,7	0,922	9,0	0,998
-1,8	0,111	0,5	0,697	2,8	0,927	9,5	0,999
-1,7	0,132	0,6	0,715	2,9	0,931	10,0	0,999
-1,6	0,156	0,7	0,732	3,0	0,935	10,5	0,999
-1,5	0,181	0,8	0,748	3,1	0,939	11,0	1,000
-1,4	0,208	0,9	0,763	3,2	0,943		
-1,3	0,235	1,0	0,777	3,3	0,946		
-1,2	0,264	1,1	0,790	3,4	0,949		
-1,1	0,293	1,2	0,803	3,5	0,952		
-1,0	0,322	1,3	0,815	3,6	0,955		
-0,9	0,351	1,4	0,826	3,7	0,958		

TABLE 2

The distribution function $\Psi(y)$

y	$\Psi(y)$	y	$\Psi(y)$	y	$\Psi(y)$	y	$\Psi(y)$
-2,0	0,000	-0,5	0,310	1,0	0,873	2,5	0,987
-1,9	0,001	-0,4	0,359	1,1	0,890	2,6	0,989
-1,8	0,002	-0,3	0,409	1,2	0,905	2,7	0,990
-1,7	0,005	-0,2	0,458	1,3	0,918	2,8	0,992
-1,6	0,009	-0,1	0,507	1,4	0,930	2,9	0,993
-1,5	0,015	0,0	0,554	1,5	0,939	3,0	0,994
-1,4	0,025	0,1	0,599	1,6	0,948	3,1	0,995
-1,3	0,038	0,2	0,641	1,7	0,955	3,2	0,996
-1,2	0,056	0,3	0,681	1,8	0,962	3,3	0,996
-1,1	0,078	0,4	0,717	1,9	0,967	3,4	0,997
-1,0	0,106	0,5	0,750	2,0	0,972	3,5	0,997
-0,9	0,139	0,6	0,780	2,1	0,976	4,0	0,999
-0,8	0,176	0,7	0,808	2,2	0,980	4,5	0,999
-0,7	0,217	0,8	0,832	2,3	0,983	5,0	1,000
-0,6	0,262	0,9	0,854	2,4	0,985		

TABLE 3

The percentiles of Ψ_0 and Ψ

α	$y_0(\alpha)$	$y(\alpha)$
0,20	1,18	0,67
0,15	1,64	0,88
0,10	2,29	1,16
0,05	3,42	1,63
0,01	6,12	2,65

TABLE 4
The constants a_n^o and a_n

n	a_n^o	a_n
1	0,79	0,00
5	1,56	0,50
10	1,86	0,67
20	2,12	0,83
30	2,25	0,91
40	2,34	0,97
50	2,40	1,01
100	2,58	1,15
200	2,73	1,27
400	2,87	1,39
600	2,94	1,46
800	2,99	1,50
1000	3,03	1,54

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